

06 · Is spend *causing* sales? — causal MMM (pymc-marketing)

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Runs in the legacy environment (pymc<6): `make env-legacy, kernel
cmp-legacy.`

The business decision. Brand search converts like crazy — should we pour budget into it? Be careful: brand search mostly **catches** demand that upper-funnel channels (TV) and seasonality already created, so crediting it for those conversions and shifting budget its way would be a mistake.

1.0.1 The concepts

- **MMM (Marketing Mix Model)** — a regression of sales on each channel’s spend over time (usually weekly), used to attribute sales to channels and set budgets. It’s the “top-down” alternative to click-tracking.
- **Adstock (carry-over)** — advertising doesn’t only work the week it runs; its effect *decays over subsequent weeks*. Adstock transforms raw spend into an “effective” exposure that carries over.
- **Saturation (diminishing returns)** — the 10th €1,000 into a channel buys less lift than the 1st. A saturation curve bends over as spend rises; its *slope* is the **marginal** return that should drive the next euro of budget.
- **Last-click attribution** — crediting the conversion to the final touch (often brand search). It systematically **over-credits** demand-capture channels, which is the error a causal MMM aims to fix.
- **Incremental vs mediated** — the *incremental* contribution of a channel is what it *caused*; a channel can show a big correlation with sales while causing very little (it was downstream of the real driver).

1.0.2 The arc of this notebook — a cautionary tale with a resolution

A plain MMM regresses sales on all channels and is fooled by a **confounder** like seasonality (it lifts sales *and* drives brand-search spend). The textbook fix is a **causal MMM**: hand it a **DAG** and let the backdoor logic (nb 05) control for the confounder. This notebook shows that fix is **necessary but not sufficient**, in three acts:

1. **The confounding correction works — in a linear model.** A naive OLS attribution that omits seasonality massively over-credits brand_search; adding seasonality as a control collapses that credit toward the small truth. That’s the cleanest way to *see* the confounding, and it’s robust.

2. **The out-of-the-box causal MMM still inverts the ranking.** Fit with library-default priors, the MMM's *flexible saturation curve* for brand_search **re-absorbs** the seasonal demand the linear control was meant to remove — so it credits brand_search *more* than TV, the opposite of the truth. A DAG plus a linear control does **not** de-confound a channel whose spend is collinear with the confounder.
3. **Experiment-calibrated priors recover it.** Feed in what a small **geo experiment** teaches — that brand_search's incremental effect is small — as an informative prior on its saturation, and the MMM ranks the channels correctly again. **This is why MMMs must be calibrated with experiments** (Anchor B, nb 07): observational data alone cannot pin the decomposition.

On real data. Swap in your **own weekly spend-by-channel + sales** table plus known confounders (seasonality, price, distribution). Public example: the datasets shipped with Meta's *Robyn* or *pymc-marketing*'s own tutorials. Always pair an MMM with occasional geo experiments to anchor the levels.

7-step contract.

1.1 2 · Simulate a ground truth

Weekly sales from a faithful **adstock + Michaelis-Menten saturation** process. **TV** is a big genuine driver. **seasonality** is a **confounder** — it lifts sales directly *and* drives brand-search spend. **brand_search** has only a *small* real effect, but its spend rides the seasonal wave (corr ~ 0.75), so a naive attribution ignoring seasonality massively over-credits it. Honest answer: TV highly incremental, brand_search only slightly.

TRUE incremental sales - TV 5,263 brand_search 518 (small)

	date_week	tv	brand_search	seasonality	sales
0	2023-01-02	42.870187	35.579322	0.000000	134.900978
1	2023-01-09	53.049712	17.732754	0.120537	137.863335
2	2023-01-16	27.623875	23.486826	0.239316	142.263680
3	2023-01-23	56.329747	34.885672	0.354605	159.788875
4	2023-01-30	43.339248	33.736743	0.464723	152.485792

1.2 3 · Identify — the marketing DAG picks controls vs mediators

DAG: seasonality -> {sales, brand_search}, tv -> sales, brand_search -> sales. The backdoor criterion says **seasonality is a confounder -> control for it**; omitting it is exactly the last-click mistake that over-credits brand search. The estimand for channel *c* is its **incremental contribution** — sum over weeks of $\mathbb{E}[\text{sales} \mid \text{do}(\text{spend}_c = \text{observed})] - \mathbb{E}[\text{sales} \mid \text{do}(\text{spend}_c = 0)]$, holding the other channels at their observed spend (how much of sales would vanish if we switched channel *c* off). ROI = contribution ÷ spend, each with a posterior. Identification comes from the backdoor set {seasonality}; we hand the DAG to the MMM via `dag`, `treatment_nodes`, `outcome_node`.

Backdoor adjustment set the causal layer will use: ['seasonality']

1.3 4 · Estimate — fit the causal MMM twice

We fit the **same DAG-aware MMM twice**, to make the lesson unmissable:

- (a) **default priors** — the honest, out-of-the-box causal MMM. Watch it *invert* the ranking.
- (b) **experiment-calibrated priors** — encode what a geo experiment teaches (brand_search’s incremental effect is small) as an informative prior on its saturation ceiling, and watch the ranking snap back.

A caution that runs through both: **MMMs are miscalibrated in *absolute* terms** — with channel spend riding seasonal demand, a short weekly series can’t fully separate “channel drove sales” from “season drove both”, so fitted contributions run high. What we’re after is the **ranking** and each channel’s **ROI verdict**.

Output()

```
(a) default-prior MMM convergence: max r-hat 1.010 - min ESS 381 - divergences
14
(a) default contribution - TV 10,390    brand_search 36,007
    true: TV 5,263, brand_search 518  ->  the MMM ranks brand_search ABOVE TV
(inverted!).
```

1.3.1 Why the out-of-the-box MMM inverts the ranking

The default MMM credits brand_search *above* TV — backwards. The mechanism is worth seeing, because it’s a real trap:

- brand_search’s fitted weekly contribution correlates **~0.8 with seasonality** — its flexible saturation curve is quietly **re-absorbing the seasonal demand** the linear seasonality control was meant to remove.
- Because brand_search rides the seasonal wave more tightly than TV does, it wins that tug-of-war and books a huge contribution (tens of thousands vs a true ~500), while a large **negative baseline** offsets it — the classic non-identified MMM level decomposition.
- The DAG was right and the control was present; they still weren’t enough. **A linear confounder-control does not de-confound a channel whose spend is collinear with the confounder and enters through a flexible functional form.** Observational data alone can’t break the tie.

1.3.2 The fix — priors calibrated from a geo experiment

The standard remedy is exactly the discipline of Anchor B (nb 07): run a **small geo experiment**, learn that brand_search’s *incremental* effect is small, and feed that back as an **informative prior on its saturation ceiling** (saturation_alpha). TV, whose incremental role we haven’t independently pinned down, keeps a weakly-informative prior. This is not putting a thumb on the scale — it’s using experimental evidence exactly where observational data is silent.

Output()

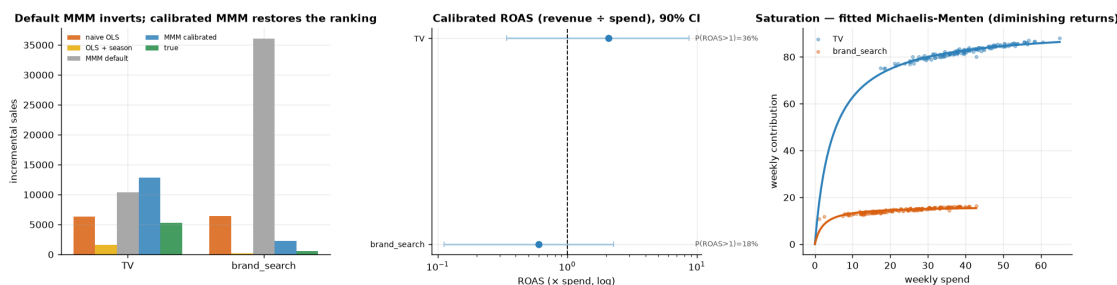
(b) calibrated-prior MMM convergence: max r-hat 1.010 - min ESS 376 - divergences 0
 (b) calibrated contribution - TV 12,820 brand_search 2,262
 true: TV 5,263, brand_search 518 -> ranking RESTORED: TV above brand_search.

1.4 5 · Validate — the correction, the recovered ROI, and saturation

Three views:

1. **The confounding correction, several ways** — naive OLS (omit season) over-credits brand_search; OLS + season collapses it toward truth; the **default** causal MMM *inverts* it; the **calibrated** MMM restores the right ranking. The contrast between the two MMMs is the headline.
2. **ROI with uncertainty (calibrated MMM)** — each channel's posterior ROI vs the break-even line. ROAS = contribution ÷ (fixed **observed** spend), so it carries the **same relative miscalibration as the contribution** — dividing by a constant spend cannot remove the bias. What is robust is the ROAS **ranking** (and brand_search's below-break-even verdict), **not** the absolute ROAS *levels*, which are biased by the same factor as the contributions.
3. **Saturation curves** — contribution vs spend per channel, showing diminishing returns (the shape that sets *marginal* ROI and hence where the next euro should go).

brand_search credit - naive 6,401 | OLS+season 222 | MMM default 36,007 | MMM calibrated 2,262 | true 518
 calibrated ROAS - TV 2.08x [90% 0.34,8.67] · brand_search 0.60x [90% 0.11,2.28]
 TRUE ROAS - TV 0.85x · brand_search 0.14x (both below 1x: the MMM recovers the ranking but overstates levels)



How to read this. *Left* tells the whole story in one panel. **Naive OLS** (orange) over-credits brand_search; **OLS + season** (gold) collapses that credit toward the small truth — the confounding correction, working. Then the two MMMs: the **default** causal MMM (grey, the failure mode) *inverts* the ranking, crediting brand_search far above TV — the seasonal-demand-stealing trap — while the **calibrated** MMM (blue) restores TV above brand_search, landing beside the **true** bars (green). *Middle* — the decision-relevant **calibrated ROI**: TV's mean clears break-even but its 90% interval is wide and dips below 1x, while brand_search's mean sits **below break-even**. So TV is the better channel, but its above-break-even verdict is itself uncertain. And the ranking is only *probably* right: draw-by-draw, TV's ROI beats brand_search's about **78%** of the time (the

$P(\text{TV ROI} > \text{brand_search ROI})$ computed next) — more likely than not, but **short of the 0.8 bar** we'd want before moving real budget. *Right* — the **saturation curves**: contribution flattens as spend rises (diminishing returns), so the *next* euro into a channel is worth its curve's *slope*, not its average. The honest caveat (Step 7): even the calibrated ROI *levels* are approximate — anchor them with the same kind of geo experiment that supplied the prior.

1.5 6 · Decide, in euros — should we reallocate budget yet?

Budget chases **incremental return**, not last-click credit. One naming point first: what the MMM gives us is **ROAS** — incremental *revenue* per euro of spend — not profit ROI. Break-even at 1x therefore assumes a 100% margin; at a realistic margin m the break-even ROAS is $1/m$ (a 30% margin $\rightarrow$ 3.3x), which we show below.

Now that experiment-calibrated priors have made the *ranking* trustworthy, the point estimates put TV above brand_search, so the *direction* of any reallocation is toward TV. But a direction is not a decision, and the **absolute levels are still miscalibrated**. So we quantify three things: (1) the sales impact of shifting a slice of budget — swept across shift sizes and made **saturation-aware**, so marginal ROAS falls as we pour more into TV; (2) the posterior probability the move actually helps; and (3) the honest gap between the MMM's ROAS and the *true* ROAS. That last check is the sobering one: it is why, even with the ranking fixed, the absolute “is TV worth it?” question still belongs to a geo test (Anchor B, nb 07), not a still-approximate MMM.

TV ROAS - average 2.08x · marginal (the next euro, at today's spend) 0.23x - marginal is lower because of saturation (diminishing returns).

Shifting 15% of brand_search spend (€566) to TV $\rightarrow$ +€61 net sales (saturation-aware); the sweep peaks near 22%.

$P(\text{TV ROAS} > \text{brand_search ROAS}) = 0.78 \rightarrow$ lean toward TV, but test first

Ranking vs levels - the watertight part:

TRUE ROAS: TV 0.85x · brand_search 0.14x (both BELOW 1x) | MMM says TV 2.08x · bs 0.60x

The MMM recovers the RANKING (TV $\gg$ brand_search) but overstates LEVELS $\sim$ 2.4x for TV.

ROAS is revenue $\div$ spend - it assumes 100% margin. At a realistic 30% margin the break-even is $1/0.30 \sim 3.3x$,

which even TV's estimated 2.08x fails; and TV's TRUE ROAS 0.85x is below break-even at ANY margin.

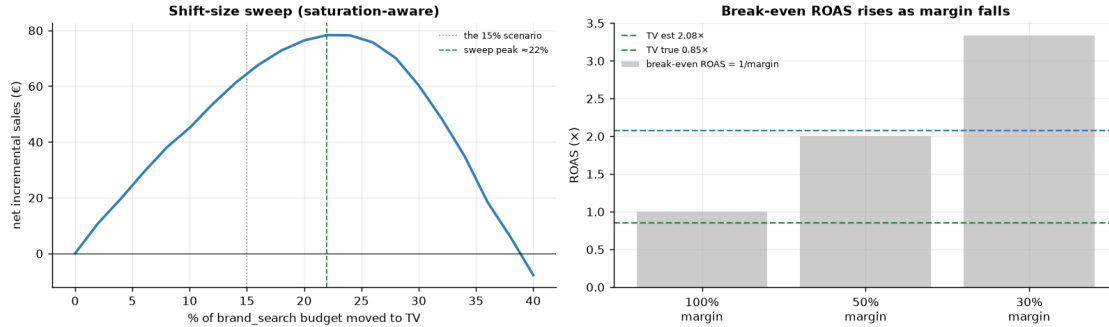
So: reallocate toward TV (the ranking is trustworthy), but 'is TV worth it in absolute €?' needs the geo

experiment (nb 07) - this MMM's absolute levels are not calibrated enough to answer it.

Decision quantity WITH uncertainty - shift 15% (€566) brand_search $\rightarrow$ TV (per-draw, average-ROAS):

Deltasales = €836 [90% €-799, €4,618] · $P(\Delta > 0) = 0.78$

TRUE Deltasales (from DGP true_contrib) = €405 $\rightarrow$ direction check: MATCH (both favour TV)



1.6 7 · Caveats

- **Absolute contributions need calibration.** Even the calibrated MMM runs a fewx high in absolute euros, because channel spend rides the seasonal wave the model can't fully net out from a short series. **Trust the ranking and the ROI verdict, not the absolute euros** — and note it took an experiment-informed prior to get even the *ranking* right. This is the single most important thing to say about any MMM.
- **A DAG is necessary but not sufficient.** The correct backdoor control (seasonality) did not stop a flexibly-saturated, collinear channel from re-absorbing the confounder. Structural assumptions still need experimental calibration to be trustworthy.
- **Adstock/saturation priors matter — a lot.** Carry-over (λ) and diminishing-returns (α , κ) are weakly identified from limited weekly data; here the saturation-ceiling prior *flipped the conclusion*. Use informative priors from experiments wherever you have them.
- **Correlated channels -> wide, correlated ROI posteriors.** When spend moves with a confounder, the data can't fully separate them; the intervals will (honestly) show it.